Latest updates: https://dl.acm.org/doi/10.1145/3746252.3761333

RESEARCH-ARTICLE

GFlowNet with Gradient-based Optimization for Bayesian Network Structure Learning

ZHU YANG, Yunnan University, Kunming, Yunnan, China

KUN YUE, Yunnan University, Kunming, Yunnan, China

ZHIWEI QI, Yunnan University, Kunming, Yunnan, China

LIANG DUAN, Yunnan University, Kunming, Yunnan, China

JIANYU LI, Yunnan University, Kunming, Yunnan, China

Open Access Support provided by:

Yunnan University

PDF Download
3746252.3761333.pdf
05 March 2026
Total Citations: 0
Total Downloads: 111

Published: 10 November 2025

Citation in BibTeX format

CIKM '25: The 34th ACM International Conference on Information and Knowledge Management
November 10 - 14, 2025
Seoul, Republic of Korea

Conference Sponsors:
SIGWEB
SIGIR

https://doi.org/10.1145/3746252.3761333
ISBN: 9798400720406

GFlowNet with Gradient-based Optimization for Bayesian Network Structure Learning

Zhu Yang
Yunnan University
School of Information Science and Engineering
Yunnan Key Laboratory of Intelligent Systems and Computing
Kunming, Yunnan, China
yangzhu@stu.ynu.edu.cn

Kun Yue
Yunnan University
School of Information Science and Engineering
Yunnan Key Laboratory of Intelligent Systems and Computing
Kunming, Yunnan, China
kyue@ynu.edu.cn

Zhiwei Qi*
Yunnan University
Yunnan Key Laboratory of Intelligent Systems and Computing
School of Education
Kunming, Yunnan, China
maryqizhiwei@ynu.edu.cn

Liang Duan
Yunnan University
School of Information Science and Engineering
Yunnan Key Laboratory of Intelligent Systems and Computing
Kunming, Yunnan, China
duanl@ynu.edu.cn

Jianyu Li
Yunnan University
School of Information Science and Engineering
Yunnan Key Laboratory of Intelligent Systems and Computing
Kunming, Yunnan, China
lijianyu@itc.ynu.edu.cn

Abstract

Bayesian network (BN) structure learning on the discrete observations is crucial for representing uncertainty in data. However, existing single-structure learning methods are commonly trapped in local optimum or yielding structures that may lead to poorly calibrated predictions. Although posterior approximation methods can quantify the epistemic uncertainty over the learned BN structures, they cannot reliably identify the optimal BN structures. To tackle these issues, we propose a GFlowNet with Gradient-based Optimization (GFlowOpt) for the BN structure learning method. Initially, we employ Bayesian Information Criterion (BIC) scores as the rewards of BN structures, and adopt a contrastive learning-based training technique on GFlowNet to efficiently generate much better DAG representations. Subsequently, we train a proxy model on the continuous DAG representations by adopting the gradient-ascent-based optimization method to search more high proxy-scoring discrete candidate DAGs. Finally, we adopt Hill Climbing (HC) on these candidate DAGs to search high-scoring DAGs as the ultimate BN structures. Extensive experiments conducted on benchmark datasets demonstrate that the superiority of our proposed method compared with other state-of-the-art methods.

CCS Concepts

• **Computing methodologies → Artificial intelligence**.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
CIKM '25, Seoul, Republic of Korea
© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 979-8-4007-2040-6/2025/11
https://doi.org/10.1145/3746252.3761333

Keywords

Bayesian network, Structure learning, Contrastive learning, Posterior approximation, Continuous optimization

ACM Reference Format:
Zhu Yang, Kun Yue, Zhiwei Qi, Liang Duan, and Jianyu Li. 2025. GFlowNet with Gradient-based Optimization for Bayesian Network Structure Learning. In *Proceedings of the 34th ACM International Conference on Information and Knowledge Management (CIKM '25), November 10–14, 2025, Seoul, Republic of Korea.* ACM, New York, NY, USA, 10 pages. https://doi.org/10.1145/3746252.3761333

1 Introduction

Bayesian network (BN) is a popular framework for representing uncertainty over a set of variables, with applications in diverse scientific domains such as medical diagnosis, molecular biology, and ecological modeling [5, 30, 32]. It is characterized by a directed acyclic graph (DAG) structure that encodes conditional dependencies among variables [24]. BN structure learning, i.e., constructing a DAG from the input data, is crucial for data-driven applications of complex uncertain systems and for discovering statistical relationships among their constituent variables. However, learning an optimal BN structure remains challenging due to the combinatorially large search space of possible structures and the constraint of graph acyclicity [27].

Most existing structure learning methods aim to recover a single DAG or its Markov equivalence class, using either score-based approaches that search over candidate structures guided by a defined scoring function [24], or constraint-based methods that perform conditional independence (CI) tests to infer the underlying structure [10]. However, these approaches typically disregard epistemic uncertainty, resulting in poorly calibrated predictions [18]. Posterior approximation methods offer a principled way to quantify

epistemic uncertainty and assess confidence in a given BN by explicitly modeling the posterior distribution over DAGs [27]. Nevertheless, such methods often suffer from limited exploration of the combinatorial structure space, which hampers their ability to consistently identify high-scoring DAGs. Therefore, it is crucial to discover the optimal DAG within each distinct high-probability region of a well-explored posterior distribution.

Generative Flow Networks (GFlowNets) [6] provide an effective framework for generating discrete composite objects, and a variant known as DAG-GFlowNet has been applied to sample DAGs from a learned posterior distribution [18]. It offers accurate approximations of the posterior, which stems from GFlowNet's ability to comprehensively explore the combinatorial structure space through diverse sampling of candidate DAGs. However, DAG-GFlowNet primarily explores high-probability regions rather than identifying precise optimal structures, meaning that the initially sampled DAGs may not include a globally optimal structure. Nevertheless, generating high-quality candidate DAGs from the posterior approximation remains valuable, particularly when using the DAG-GFlowNet-sampled candidates as a starting point for further refinement or more targeted generation strategies.

Continuous representation optimization has emerged as an effective approach for optimizing discrete structures through gradient-based updates [28], enabling more efficient structural encoding and offering simpler and smoother optimization paths compared to direct discrete search. Motivated by this, we propose to map the discrete candidate DAGs generated by DAG-GFlowNet into a continuous representation space in order to identify optimal DAGs. Accordingly, we aim to address the following two challenges:

(a) How to map the discrete DAGs into a continuous representation space, yielding representations that are highly conducive to optimization?
(b) How to obtain high-scoring DAGs by optimizing the continuous representations based on the gradients?

To map DAGs into a continuous representation space, we propose a structure sampling method that leverages the inherent graph encoding capabilities of DAG-GFlowNet to generate candidate DAGs along with their corresponding representations. To facilitate the GFlowNet's ability to approximate the posterior distribution over DAGs, we employ the Bayesian Information Criterion (BIC) score as a reward signal during training, encouraging the model to generate DAGs with probabilities proportional to their posterior estimates. Furthermore, to ensure that the continuous representations produced by GFlowNet are amenable to subsequent optimization, we incorporate a contrastive learning (CL) objective into the training process. This helps refine the discriminative quality and structural fidelity of the learned representations.

To identify optimal DAGs, we propose a two-stage optimization strategy that operates on the continuous representations of DAGs in order to discover high-scoring structures. In the first stage, a proxy model is trained to learn a continuous and differentiable score landscape from the representations generated by GFlowNet. This learned landscape guides a gradient ascent procedure in the continuous space, enabling the identification of optimized representations corresponding to regions of high predicted scores. These refined representations are then used to select more promising DAGs from the candidate set produced by GFlowNet. In the second stage, the selected DAGs serve as initial structures for a Hill Climbing (HC) algorithm, which performs local search in the discrete DAG space to further refine and identify the optimal structures.

Our main contributions are summarized as follows:

- We propose a GFlowNet-based structure sampling method that generates diverse candidate DAGs along with continuous representations suitable for gradient-based optimization.
- We introduce a novel two-stage continuous-to-discrete optimization strategy that efficiently identifies high-scoring BN structures.
- Extensive experiments on nine benchmark datasets show that our method significantly outperforms state-of-the-art approaches in discovering high-quality BN structures.

2 Related Work

Single Structure Discovery Methods. These methods produce a single structure using score-based, constraint-based, or hybrid approaches [23]. *Score-based methods* aim to maximize a score function [9, 11], commonly employing heuristic searches like Hill Climbing (HC) [22]. Such searches are often enhanced with techniques like Tabu search [7] for navigating local optima, or integrated into broader strategies such as Model Averaging Hill Climbing (MAHC) [17] for improved robustness. *Constraint-based methods* infer structure from CI tests [24], with notable algorithms like PC [36] and its variant PC-Stable [12]. *Hybrid approaches* merge these strategies [34], typically using constraints to prune the search space for score-based algorithms; examples include Max-Min Hill Climbing (MMHC) [38] and H2PC [21]. However, single-structure methods reliance on heuristic searches makes them prone to converging to local optima, potentially overlooking higher-scoring structures due to an incomplete exploration of the vast search space.

Posterior Approximation Methods. Different from single structure methods, posterior approximation methods estimate the structure distribution over structures from observations, primarily via Markov Chain Monte Carlo (MCMC), Variational Inference (VI), and GFlowNet. *MCMC methods* sample from the posterior DAG distribution [29]; techniques to improve mixing include order sampling [20], specialized sample spaces and moves [25], yielding efficient samplers such as Gadget [39]. *VI methods* approximate the joint posterior [3], often using continuous relaxations of the discrete DAG space [27]. *GFlowNet* provide another paradigm for sampling BNs, e.g., DAG-GFlowNet [18] for the marginal posterior over structures [6], and JSP-GFN [19] for the joint posterior of structures and parameters. However, these approaches may insufficiently explore identified high-probability modes, potentially hindering convergence to the single highest-scoring structure within them.

3 Preliminary

Definition 1. A BN provides a probabilistic graphical model for a set of m random variables $\{X_1, \ldots, X_m\}$ of two components: structure and parameters. The structure is represented by a DAG $G = (X, E)$, where V is the set of nodes representing the variables $\{X_1, \ldots, X_m\}$ and $E \subseteq X \times X$ is the set of directed edges indicating conditional dependencies. We denote a directed edge from X_i to X_j as $(X_i, X_j) \in E$ or $X_i \rightarrow X_j \in G$. The set of parents for a node

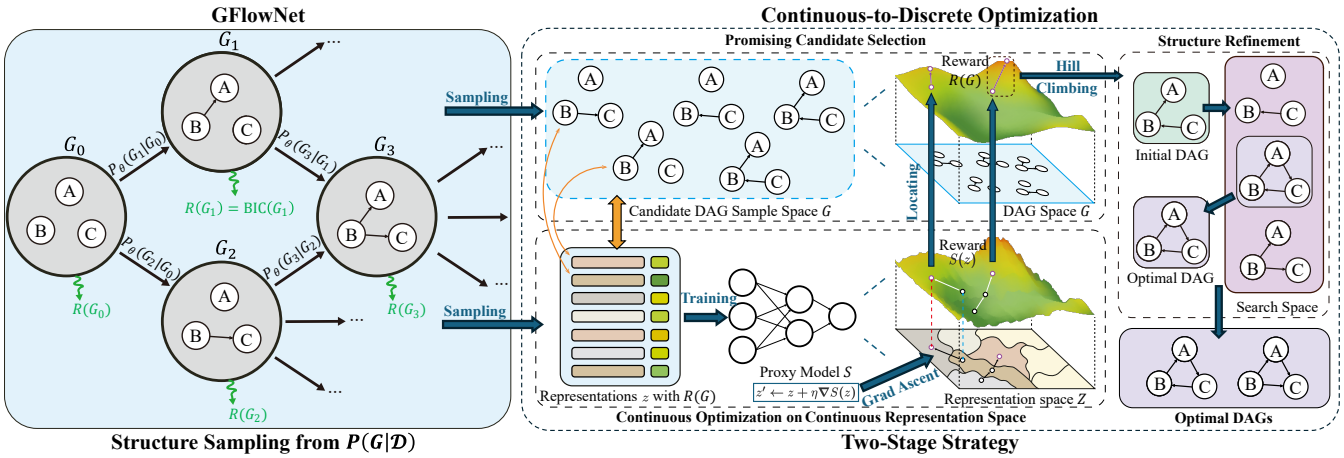

Figure 1: The framework of our BN structure discovery model consists of three components: (a) a GFlowNet-based structure sampling module for generating candidate DAGs; (b) a continuous optimization module that trains a proxy model S on the sampled structures and identifies the most promising DAGs through gradient ascent on S's continuous landscape; and (c) a structure refinement module that takes the obtained DAGs as initialization and applies Hill Climbing refinement to find the optimal DAGs.

X_i in G is denoted $\mathrm{Pa}_G(X_i)$. G implies a factorization of the joint probability distribution as:

$$P(X_1, \ldots, X_m \mid \theta, G) = \prod_{i=1}^{m} P(X_i \mid \mathrm{Pa}_G(X_i); \theta_i), \tag{1}$$

where $\theta = \{\theta_1, \ldots, \theta_n\}$ represents the parameters governing the conditional probability distributions (CPDs) $P(X_i \mid \mathrm{Pa}_G(X_i))$. Given a dataset of observations $\mathcal{D}$ if the structure G is known, learning typically involves estimating the parameters θ [24]. Alternatively, if the DAG structure G is unknown, the focus shifts to structure learning: the task of inferring G from the data $\mathcal{D}$ to uncover the conditional dependencies it represents [37].

Definition 2. A GFlowNet operates over a state space $\mathcal{S}$ structured as a DAG G, where the target sample space $\mathcal{X}$ is a subset of states ($\mathcal{X} \subseteq \mathcal{S}$). Samples $x \in \mathcal{X}$, called complete states, are constructed via sequential transitions along the edges of G, starting from a fixed initial state s_0. A terminal state s_f often marks the end of a construction sequence, forming a complete trajectory τ from s_0 to s_f via a state $x \in \mathcal{X}$. The core mechanism involves learning a non-negative flow function $F_\theta(s \to s')$ over the DAG edges, typically parameterized by θ and is trained to satisfy the flow-matching conditions at each state $s' \neq s_0$:

$$\sum_{s \in \mathrm{Pa}(s')} F_\theta(s \to s') - \sum_{s'' \in \mathrm{Ch}(s')} F_\theta(s' \to s'') = R(s'). \tag{2}$$

This condition essentially balances the incoming flow and the outgoing flow with the reward at the state. Satisfying Eq. (2) allows the GFlowNet to induce a generative process to sample complete states $x \in \mathcal{X}$ with probability proportional to $R(x)$. Following a forward transition probability, defined as:

$$P_\theta(s_{t+1} | s_t) \propto F_\theta(s_t \to s_{t+1}). \tag{3}$$

Sampling trajectories according to P_θ guarantees that the marginal probability of generating a complete state x is proportional to its reward: $P(x) \propto R(x)$. Crucially, unlike reinforcement learning methods focused on maximizing rewards, GFlowNet aim to model the entire distribution over $\mathcal{X}$ proportional to the rewards $R(x)$.

Problem Formalization. We aim to discover a set of high-score DAGs within region discovered by GFlowNet of the posterior $P(G \mid \mathcal{D})$. We first employ GFlowNet, trained with reward $R(G) \propto P(\mathcal{D} \mid G)P(G)$ and contrastive representation learning, to explore the posterior and learn informative structure representations z. Our framework then performs continuous optimization: a proxy model S is trained to define a differentiable approximation of the score landscape, which guides gradient ascent in the continuous space to identify optimized representations that are subsequently mapped to promising discrete candidate DAGs, G_c. Finally, these promising DAGs G_c undergo a discrete local search to pinpoint locally optimal, high-scoring DAGs, G^*.

4 Methodology

4.1 Framework Overview

Figure 1 illustrates the framework of our Bayesian Network (BN) structure discovery model, GFlowOpt, which generates high-quality DAG structures through the following steps:

- **GFlowNet Structure Sampling** first integrates the BIC score and contrastive learning techniques to learn a posterior distribution from the input data, and then generates candidate DAGs along with their corresponding representations by sampling from the learned distribution.
- **Two-stage Optimization Strategy** characterizes the representations of candidate DAGs via a proxy model and identifies the most promising DAGs through gradient ascent on the

model. Subsequently, a BIC-based Hill Climbing algorithm is employed to determine the optimal DAGs.

4.2 GFlowNet Structure Sampling

4.2.1 BIC-based GFlowNet Reward. To learn an approximate posterior distribution $P(G \mid \mathcal{D})$ over DAGs, we utilize the DAG-GFlowNet [18] to learn the forward transition policy for generating a DAG G that yields sampling probabilities $P(G)$ proportional to a specified reward. The forward transition policy training relies on minimizing a specific objective function, as follows:

$$\mathcal{L}_{DB}(\theta) = \mathbb{E}_{(G \to G') \sim \pi}\left[\left(\log \frac{R(G')P_B(G|G')P_\theta(s_f|G)}{R(G)P_\theta(G'|G)P_\theta(s_f|G')}\right)^2\right], \quad (4)$$

where $G \to G'$ represents adding an edge between G and G', $P_\theta(G'|G)$ is the learnable forward transition policy, $P_\theta(s_f|G)$ is the termination probability at state G, $P_B(G|G')$ denotes a fixed backward transition policy, uniform over parent states, and π is the transition sampling distribution. $R(G)$ is the reward function and is defined as follows:

$$R(G) = P(G)P(\mathcal{D}|G), \quad (5)$$

where $P(G)$ is the prior probability of the structure G, and $P(\mathcal{D}|G)$ is the marginal likelihood of the data given G.

To efficiently calculate reward, we adopt the BIC score metric to evaluate $R(G)$ rather than the Bayesian Dirichlet equivalent (BDe) score commonly used in DAG-GFlowNet, in order to avoid the substantial computational burden incurred by the BDe score's reliance on Gamma functions [24]. The BIC score is defined as follows:

$$\mathrm{BIC}(G, \mathcal{D}) = \mathcal{L}(G, \mathcal{D}) - \frac{k_G}{2} \log n, \quad (6)$$

where $\mathcal{L}(G, \mathcal{D})$ is the maximized log-likelihood of the data $\mathcal{D}$ for the given graph structure G, k_G represents the number of parameters in the model associated with graph G, and n is the total number of samples or observations in the dataset $\mathcal{D}$.

To ensure that the reward mechanism in GFlowNet effectively approximates the posterior distribution of DAGs, we propose a BIC-based reward calculation method.

First, we establish the connection between the BIC score and the posterior probability $P(G|\mathcal{D})$. The log-posterior probability of a graph structure G given data $\mathcal{D}$ is expressed by Bayes' theorem as:

$$\log P(G|\mathcal{D}) = \log P(\mathcal{D}|G) + \log P(G) - \log P(\mathcal{D}). \quad (7)$$

Note that BIC score serves as an approximation to the log marginal likelihood $\log P(\mathcal{D}|G)$, particularly when the number of observations is sufficiently large [24], we have $\mathrm{BIC}(G, \mathcal{D}) \approx \log P(\mathcal{D}|G)$. Furthermore, a uniform prior over BN structures $P(G)$ being constant makes $\log P(G)$ a constant. Substituting the assumption and the BIC approximation into Eq. (7), we have

$$\log P(G|\mathcal{D}) \approx \mathrm{BIC}(G, \mathcal{D}) + C, \quad (8)$$

where C is a constant incorporating $\log P(G) - \log P(\mathcal{D})$.

Exponentiating both sides shows the relationship $P(G|\mathcal{D}) \propto \exp(\mathrm{BIC}(G, \mathcal{D}))$. Therefore, by defining the GFlowNet reward as $R(G) = \exp(\mathrm{BIC}(G, \mathcal{D}))$, GFlowNet is guided to learn an agent that samples structures G with probabilities proportional to $P(G|\mathcal{D})$, thereby approximating the target posterior distribution.

Moreover, the BIC score satisfies the modularity property, a crucial characteristic for the efficient training of GFlowNet. This property allows that $\mathrm{BIC}(G, \mathcal{D})$ to be decomposed as a sum of local scores involving each of the m variables X_i and its respective parent set $Pa_G(X_i)$ [24]:

$$\mathrm{BIC}(G, \mathcal{D}) = \sum_{i=1}^{m} \mathrm{BIC}_{\mathrm{local}}(X_i, Pa_G(X_i), \mathcal{D}). \quad (9)$$

Example 1. *The GFlowNet training objective (detailed balance loss as in Eq. (4)) often relies on calculating the log-reward ratio, or delta score,* $\log R(G') - \log R(G)$*, between consecutive states* G *and* G'*.*

By using $R(G) = \exp(\mathrm{BIC}(G, \mathcal{D}))$, the modularity of the BIC score (Eq. (9)) enables efficient calculation of this delta score. Specifically, for an edge addition $X_j \to X_i$ that transforms graph G to G', this delta score simplifies to the change in X_i's local BIC score contribution:

$$\begin{aligned}\log R(G') - \log R(G) = {} & \mathrm{BIC}_{\mathrm{local}}(X_i, Pa_{G'}(X_i), \mathcal{D}) \\ & - \mathrm{BIC}_{\mathrm{local}}(X_i, Pa_G(X_i), \mathcal{D}).\end{aligned} \quad (10)$$

4.2.2 DAG mapping. Based on the R_G, we adopt the forward transition policy P_θ from DAG-GFlowNet to learn an approximation of the posterior distribution. Given an initial graph without edges, P_θ constructs the DAG by deciding whether to transition G to G' or to output the current DAG at each construction step. The probability of transitioning G to G' by adding an edge is calculated as follows:

$$P_\theta(G'|G) = (1 - P_\theta(\mathrm{stop}|G))P_\theta(G'|G, \neg\mathrm{stop}), \quad (11)$$

where $P_\theta(\mathrm{stop}|G)$ denotes the probability of terminating the graph construction. $P_\theta(G'|G, \neg\mathrm{stop})$ denotes the probability of transitioning G to G'.

$P_\theta(\mathrm{stop}|G)$ and $P_\theta(G'|G, \neg\mathrm{stop})$ are conditioned on representations of the current graph G, which means that developing high-quality representations of G is essential for effective policy learning. Thus, we utilize GNN [4] and Self-Attention to encode G as follows:

$$z, \{u_k, v_k\}_{k=1}^{m} = \mathrm{SelfAttention}_\theta(\mathrm{GNN}_\theta(G)). \quad (12)$$

where m denotes the number of nodes of G, z denotes d-dimensional representation of G, u_k and v_k denote the representations of the k-th node when acting as a source or target node, respectively.

Then, we employ z to decide whether the process of DAG construction should be terminated or not, and node representations u_k and v_k to decide on subsequent edge additions. Specifically, $P_\theta(\mathrm{stop} \mid G)$ is estimated using a sigmoid function, while $P_\theta(G' \mid G, \neg\mathrm{stop})$ for transitioning G to G' is estimated as follows:

$$P_\theta(G' \mid G, \neg\mathrm{stop}) = \mathrm{Softmax}(A_{ij} \exp(u_i^T v_j)), \quad (13)$$

where u_i and v_j are the representations of the source and target nodes, respectively.

Note that G' is derived from G by adding a single edge. This implies that closer representations of G' and G enable smoother structural transitions in the continuous representation space. Thus, we incorporate a CL objective to improve the global graph representations z, enabling the optimization-ready representation to be

generated by P_θ. To this end, given an anchor graph G_t with its successor G_{t+1} within the same DAG-GFlowNet trajectory τ, we treat G_{t+1} as a positive sample of G_t, and randomly generate negative samples from other trajectories. We employ the following NT-Xent loss [41] to maximize the representations similarity between G' and G:

$$\mathcal{L}_{CL}^{(i)} = -\log \frac{\exp(\mathrm{sim}(\hat{z}_t^{(i)}, \hat{z}_{t+1}^{(i)})/T)}{\sum_{j=1, j\neq i}^{B} \exp(\mathrm{sim}(\hat{z}_t^{(i)}, \hat{z}_{t+1}^{(j)})/T)}, \tag{14}$$

where $\mathrm{sim}(\hat{z}_t^{(i)}, \hat{z}_{t+1}^{(i)})$ is cosine similarity (as $\hat{z}$ are L2-normalized), and T is a temperature hyperparameter.

The sum in the denominator is over the $B-1$ in-batch negative samples for $\hat{z}_t^{(i)}$ (i.e., $\hat{z}_{t+1}^{(j)}$ where $j \neq i$). The final contrastive loss for the batch, $\mathcal{L}_{CL}$, is the average of these individual losses:

$$\mathcal{L}_{CL} = \frac{1}{B} \sum_{i=1}^{B} \mathcal{L}_{CL}^{(i)}. \tag{15}$$

Example 2. *As illustrated in Figure 2, consider the transition $G_1 \longrightarrow G_2$ within trajectory τ_0. G_1 is the anchor graph, and G_2 serves as the positive sample of G_1. In contrast, G_6 in τ_1 and G_4 in τ_B are negative samples that are randomly selected from the trajectories except τ_0 within the same batch.*

To ensure the representations of G_t and G_{t+1} are as similar as possible in the transition process, we introduce a scoring function $f(z_a, z_b)$ for the temperature-scaled similarity between L2-normalized global graph representations z_a and z_b to formulate $\mathcal{L}_{CL}$. That is:

$$\begin{aligned}
\mathcal{L}_{CL} &= -\frac{1}{B} \sum_{i=1}^{B} \log \frac{\exp(f(z_t^{(i)}, z_{t+1}^{(i)}))}{\sum_{j=1, i\neq j}^{B} \exp(f(z_t^{(i)}, z_{t+1}^{(j)}))} \\
&= -\frac{1}{B} \sum_{i=1}^{B} f(z_t^{(i)}, z_{t+1}^{(i)}) \\
&\quad + \frac{1}{B} \sum_{i=1}^{B} \log \left(\sum_{j=1, i\neq j}^{B} \exp(f(z_t^{(i)}, z_{t+1}^{(j)})) \right) \\
&= -\frac{1}{B} \sum_{i=1}^{B} f(z_t^{(i)}, z_{t+1}^{(i)}) + \log B \\
&\quad + \frac{1}{B} \sum_{i=1}^{B} \log \left(\frac{1}{B} \sum_{j=1, i\neq j}^{B} \exp(f(z_t^{(i)}, z_{t+1}^{(j)})) \right).
\end{aligned} \tag{16}$$

The average summations presented in Eq. (16) can be expressed as expectations:

$$\begin{aligned}
\mathcal{L}_{CL} &= -\mathbb{E}_{P(z_t, z_{t+1})} \left[f(z_t, z_{t+1}) \right] + \log B \\
&\quad + \mathbb{E}_{P(z_t)} \log \left\{ \mathbb{E}_{P(z_{t+1})} \left[\exp(f(z_t, z_{t+1})) \right] \right\} \\
&= \mathbb{E}_{P(z_t)} \Bigg\{ -\mathbb{E}_{P(z_{t+1}|z_t)} \left[f(z_t, z_{t+1}) \right] \\
&\quad + \log \left\{ \mathbb{E}_{P(z_{t+1})} \left[\exp(f(z_t, z_{t+1})) \right] \right\} \Bigg\} + \log B,
\end{aligned} \tag{17}$$

where $P(z_t)$ is the distribution of the anchor graph, $P(z_{t+1} \mid z_t)$ is the distribution for positive pairs, and $P(z_{t+1})$ is the distribution of negative samples.

Based on Eq. (17), positive samples are drawn from $P(z_{t+1} \mid z_t)$. This distribution over representations is determined by $P_\theta(G_{t+1} \mid G_t)$. Consequently, by training $P_\theta(G_t \mid G_{t+1})$, we can generate the positive samples.

To effectively harness the representational benefits from CL alongside the GFlowNet's primary training via the detailed-balance loss, we jointly train $\mathcal{L}_{CL}$ and $\mathcal{L}_{DB}$ as follows:

$$\mathcal{L}_{\mathrm{t}} = \mathcal{L}_{DB} + \lambda \mathcal{L}_{CL}, \tag{18}$$

where λ is a hyperparameter weighting the contribution of the contrastive loss.

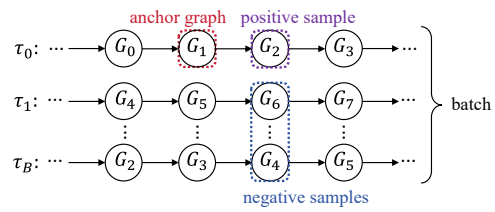

Figure 2: Contrastive loss samples in the same batch.

In the training process of GFlowNet structure sampling, we sample candidate DAGs by $P_\theta(G_{t+1} \mid G_t)$. When updating θ, the gradient is calculated by Eq. (18). The above idea is summarized in Algorithm 1. Suppose a DAG with m variables and d-dimensional representations; the cost of sampling DAGs in step 3 takes $O(m^2 \times d)$ time. In the worst case, Algorithm (1) requires processing all iterations before convergence, and the time complexity is $O(m^2 \times d \times \mathcal{N})$, where $\mathcal{N}$ is the number of iteration.

Algorithm 1 Training GFlowNet structure Sampling

Input: $\mathcal{D}$: observations, η_G: GFlowNet learning rate; ϵ_{g}: convergence threshold;
Output: Trained P_θ

1: Initialize P_θ parameters θ
2: **repeat**
3: Sample batch B of trajectories using $P_\theta(G_{t+1} \mid G_t)$
4: Evaluate total loss $\mathcal{L}_{\mathrm{t}}(\theta)$ on B ▷ (18)
5: $\theta \leftarrow \theta - \eta_G \nabla_\theta \mathcal{L}_{\mathrm{t}}(\theta)$
6: **until** $\mathcal{L}_{\mathrm{t}}(\theta) < \epsilon_{\mathrm{g}}$
7: **Return** θ

4.3 Two-stage Strategy

4.3.1 Continuous Optimization Strategy. Based on the learned policy P_θ from the structure sampling phase, the candidate DAG G along with reward $R(G)$ and representation z can be generated.

To obtain the optimal z corresponding to regions of high predicted score, we propose a proxy model to guide the continuous optimization, which learns a differentiable approximation of the

BIC score landscape from G. The proxy model is an MLP that processes input z via a configurable series of hidden layers and a final linear output layer producing the scalar score $S(z)$, as follows:

$$\begin{cases} h^{(0)} = z \\ h^{(l)} = \mathrm{ReLU}\left(\mathbf{W}^{(l)} h^{(l-1)} + \mathbf{b}^{(l)}\right), \quad \text{for } l = 1, \ldots, L \\ S(z) = \mathbf{W} h^{(L)} + b^{(\mathrm{L})} \end{cases} \tag{19}$$

where $\mathbf{W}$ and $\mathbf{b}$ are the trainable weights and biases, respectively. L denotes the number of hidden layers.

We calculate the loss function by minimizing the following mean squared error (MSE) between $S(z_i)$ and $R(G_i)$ as follows:

$$\mathcal{L}_{\mathrm{p}} = \frac{1}{B_p} \sum_{i=1}^{B_P} (S(z_i) - R(G_i))^2, \tag{20}$$

where B_p is the size of the proxy's training set; the trained model $S(z)$ thereby provides the differentiable score landscape essential for subsequent gradient-based exploration in the continuous space.

To obtain z that yield high scores w.r.t. $S(z)$, We optimize z via gradient ascent, as follows:

$$z_{k+1} = z_k + \eta \nabla_{z_k} S(z_k), \tag{21}$$

where z_k is the k-th iteration's representation and η is the step size.

The process of gradient ascent yields representation z^*, which is mapped to a promising DAG G_c. We calculate the similarity between z^* and each z derived from the set of DAGs sampled by P_θ. Formally,

$$j^* = \underset{j \in \{1,\ldots,N_{\mathrm{p}}\}}{\arg\max} \ \mathrm{sim}(z^*, z_j), \tag{22}$$

$$G_c(z^*) = G_{j^*}, \tag{23}$$

where N_{p} is the number of DAG representations sampled by P_θ.

4.3.2 Discrete Optimization Strategy. To obtain the optimal DAG G^*, we employ the HC algorithm to refine each G_c to a local optimum based on the BIC score. To this end, it explores the neighborhood $N(G_c)$ of G_c which comprises structures accessible via single edge modifications such as additions, deletions, or reversals. HC algorithm above idea is summarized G' if it offers an improved BIC score, as follows:

$$G_c \leftarrow \begin{cases} G' & \text{if } \mathrm{BIC}(G', \mathcal{D}) > \mathrm{BIC}(G_c, \mathcal{D}) \\ G_c & \text{otherwise} \end{cases} \tag{24}$$

The iterative refinement process terminates when no neighboring structure modification yields an improved BIC score, consequently converging to a locally optimal structure G^*. This hybrid optimization approach integrates proxy-guided continuous exploration in the continuous domain with precise discrete optimization, thereby establishing a framework for discovering high-quality DAGs. The above idea is summarized in Algorithm 2. Let $O(T_p)$ denote calculating the loss by Eq. 20, respectively. In the worst case, the Algorithm 2 requires processing all training samples before convergence, and the time complexity is $O(T_p \times |\mathcal{D}|)$.

Algorithm 2 Two-stage Optimization Process

Input: $G, R(G), z$; η_P: Proxy model learning rate; ϵ_P: Proxy convergence threshold;
Output: G^*: optimized structures

```
Initialize Proxy model S parameters θ_S
Create training pairs D_t = {(z, R(G))}
repeat
    Sample batch B_P from D_t
    Evaluate proxy MSE loss L_p(θ_S) on B_P          ▷ (20)
    θ_S ← θ_S − η_P ∇_{θ_S} L_p(θ_S)
until L_p(θ_S) < ε_P
Optimize z via gradient ascent on proxy S to yield z*   ▷ (21)
Select candidates G_c by similarity of z* to z          ▷ (22)
Refine G_c via HC to find optimal G*                    ▷ (24)
Return G*
```

5 Experiments

5.1 Experiment Setup

Datasets. We evaluate our method primarily on synthetic datasets generated from nine diverse benchmark BN structures. For each, we generated a dataset comprising 5000 observations via forward sampling. These benchmark networks are detailed below:

- **Asia:** A compact network for clinical patient diagnosis [26].
- **Sachs:** An empirical protein signaling network for causal discovery evaluation [32].
- **Sports:** A small BN predicting football match outcomes using team ratings and performance data [14].
- **Child:** A well-established belief network for pediatric diagnosis of congenital heart diseases [35].
- **Alarm:** A medium network for an ICU patient monitoring alarm system [5].
- **Property:** A medium BN assessing UK property market investment decisions [15].
- **Hailfinder:** A large network for forecasting severe weather in northeastern Colorado [1].
- **Win95pts:** A large, discrete network for Windows 95 printer troubleshooting [8].
- **ForMed:** A large BN modeling reoffending risk and interventions for mentally ill prisoners [16].

Structural properties of these benchmark networks are summarized in Table 1.

Comparison Methods. We adopted the following prominent structure learning methods for comparison:

- **Hill Climbing (HC):** A score-based local search that iteratively makes single arc changes to maximize a score [22].
- **Model Averaging Hill Climbing(MAHC):** An enhanced score-based method incorporating model averaging into HC for improved robustness and escaping local optima. [17].
- **PC-MAX:** A constraint-based PC algorithm variant that uses CI tests to build the graph skeleton, with enhancements for test reliability and edge orientation [31].
- **Hybrid HPC (H2PC):** A hybrid approach using constraint-based techniques to identify candidate parent/child sets, followed by a score-based search within this limited space [21].

Table 1: Properties of benchmark BN datasets.

Dataset	Nodes	Arcs	Free Parameters
Asia	8	8	18
Sachs	11	17	178
Sports	9	15	1049
Child	20	25	230
Alarm	37	46	509
Property	27	31	3056
Hailfinder	56	26	2656
Win95pts	76	112	574
ForMed	88	138	912

- **DAG-GFlowNet:** A posterior approximation method using GFlowNet to learn a policy for constructing DAGs, allowing sampling from a learned posterior over structures [18].

Metrics. We evaluate the effectiveness of the methods using the following metrics:

- **BIC Score** measures a DAG's fit to observational data while penalizing model complexity; higher scores indicate preferred structures [9].
- **Structural Hamming Distance (SHD)**: Counts edge differences between a learned DAG and the ground truth; lower SHD signifies greater structural accuracy [38].

Environment. Baseline implementations included HC (`pgmpy` [2]), MAHC (`Bayesys` [13]), PC-MAX (`Tetrad` [40]), H2PC (`bnlearn` [33]), and DAG-GFlowNet (official source code [18]). Our GflowOpt method was implemented using Python 3.9. To account for stochasticity, relevant methods all experiments were run 10 times per dataset with different random seeds. All our codes are publicly available[1].

5.2 Experimental Results

5.2.1 Effectiveness. We evaluate our method's effectiveness against baselines by assessing its ability to discover high-scoring structures, the structural accuracy of those structures relative to ground truth, and the added value of its refinement stage compared to the base DAG-GFlowNet sampler.

Comparison of Achieved Scores. Figure 3 displays the maximum BIC score distributions achieved by GFlowOpt against key baselines (HC, MAHC, H2PC) across nine benchmark datasets; other methods like PC-MAX and DAG-GFlowNet are omitted due to their significantly lower scores. Our findings indicate that:

(a) GflowOpt generally yields higher max BIC scores with lower variability than baselines on most datasets .

(b) It demonstrates notable average BIC score improvements over second-best baselines, including +252.48 points (0.46%) on Alarm (vs. HC), +202.91 points (0.43%) on Win95pts (vs. HC), and +387.63 points (0.18%) on ForMed (vs. H2PC).

(c) BIC score advantage for GflowOpt over the next-best baseline typically scales with network size, with improvements ranging from +6.00 points (0.05%) on the small Asia dataset to +387.63 points (0.18%) on ForMed.

[1]https://github.com/yy444/GflowOpt

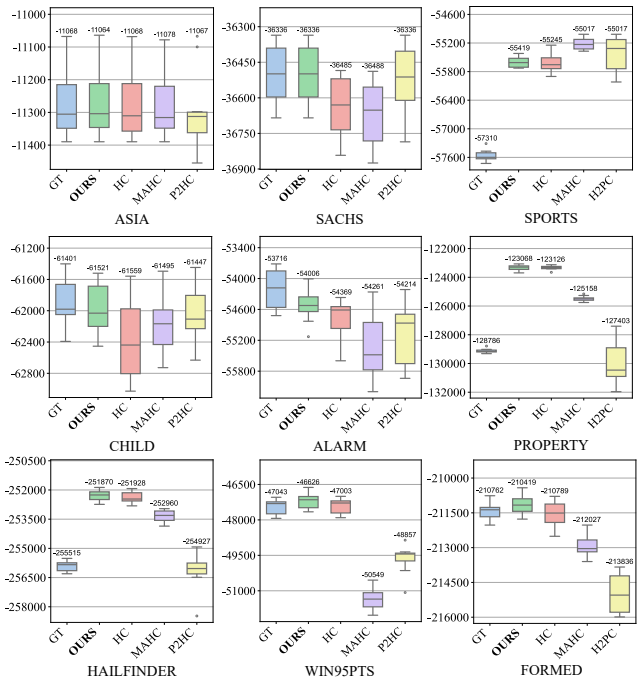

Figure 3: BIC score distributions for GFlowOpt (OURS), baselines (HC, MAHC, H2PC), and ground truth (GT) on nine benchmark datasets (N=5000; per-dataset panels). Vertical axis: BIC score; Horizontal axis: methods/GT.

Collectively, these results underscore GFlowOpt's improved stability and effectiveness in identifying structures with higher marginal likelihoods, especially as network complexity increases. For the Sports dataset, however, this suggests that the posterior modes initially discovered by GFlowNet may not always align with regions containing the ultimately highest-scoring structures.

Evaluation of Structural Accuracy. Table 2 summarizes the average SHD to the ground truth for the highest BIC-scoring structures from GFlowOpt and baselines. We can find that:

(a) GFlowOpt achieves the lowest SHD on a majority (5 out of 9) of datasets, including Asia, Sachs, Child, Property, and Hailfinder. Notably, its best SHD was 1.7 (on Asia), and its largest margin was on Hailfinder (40.6 vs. PC-MAX's 47.8, a 7.2-point improvement).

(b) On the remaining four datasets, other baselines recorded lower SHDs: PC-MAX on Alarm (16.4), H2PC on Sports (6.1) and ForMed (69.8), and MAHC on Win95pts (69.3). In these instances, GFlowOpt's SHD was higher, trailing the best by 2.5 points (ForMed) to 22.1 points (Win95pts).

These findings suggest that GflowOpt's primary objective of maximizing BIC scores frequently aligns with high structural accuracy. However, a trade-off with achieving the absolute minimum SHD is apparent in some instances, indicating that optimizing for BIC score does not always equate to optimizing directly for structural fidelity against a ground truth.

Table 2: Comparison of average SHD to ground truth for the highest BIC-scoring structures discovered by GFlowOpt and baseline methods across benchmark datasets. Best average score per column is bolded.

Methods	Small Networks			Medium Networks			Large Networks		
	Asia	Sachs	Sports	Child	Alarm	Property	Hailfinder	Win95pts	ForMed
HillClimb	6.8±2.1	9.4±2.0	15.0±4.4	9.7±3.7	29.5±5.1	17.7±1.5	49.8±9.3	107.1±8.8	88.7±16.3
MAHC	2.6±0.6	10.7±0.6	14.8±0.2	4.9±1.5	19.4±1.5	31.1±0.0	50.0±17.6	**69.3±6.0**	138.1±6.0
H2PC	6.7±1.2	8.2±0.6	**6.1±0.1**	9.2±1.2	21.4±0.6	16.9±1.7	51.6±0.6	70.0±1.7	**69.8±2.1**
PC-MAX	6.3±0.6	10.4±1.5	13.0±0.0	13.2±4.0	**16.4±2.1**	16.0±1.7	47.8±7.1	70.4±1.5	99.3±0.6
GFlowOpt	**1.7±0.7**	**4.9±1.5**	9.3±0.6	**4.8±2.7**	27.7±5.0	**15.0±4.6**	**40.6±5.6**	91.4±7.6	72.3±2.1

Evaluation of Optimization Stage. To assess the impact of its optimization stage, GflowOpt is compared to the baseline DAG-GFlowNet [18]. Table 3 details this comparison on nine datasets using Top-1 BIC and Avg Top-10 BIC scores. Key findings include:

(a) GFlowOpt consistently outperforms DAG-GFlowNet on all nine datasets for both metrics, and this improvement scales with network size.

(b) For instance, Top-1 BIC gains range from 165 points (1.4% on Asia) to 96,340 (27.6% on Hailfinder), with similar trends for Avg Top-10 BIC.

(c) GFlowOpt also generally exhibits smaller BIC score standard deviations than DAG-GFlowNet on most datasets. For Top-1 BIC, score variability reductions range from 43 (Asia) to 463 (Win95pts).

These results quantify the significant benefit of the optimization stage, demonstrating GFlowOpt's ability to refine initial structures from its approximated posterior $P(\mathcal{D} \mid G)$ to consistently discover networks with superior BIC scores, particularly for larger, more complex networks.

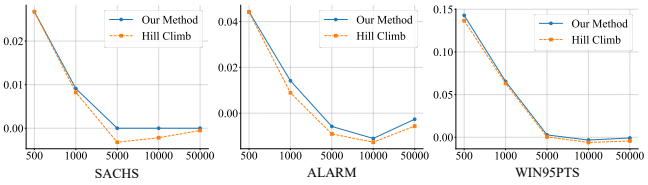

Figure 4: Scalability of GflowOpt vs. HC on Sachs, Alarm, and Win95pts networks. Horizontal axis: sample size (N); Vertical axis: Normalized BIC Score Difference from ground truth. Positive values indicate method scores exceed ground truth.

5.2.2 Scalability. Figure 4 compares GflowOpt against the HC baseline on Sachs, Alarm, and Win95pts networks across varying sample sizes N, using the Normalized BIC Difference—the normalized result of subtracting the ground truth structure's BIC score from that of the learned structure. Key observations are:

(a) GflowOpt consistently achieves a higher Normalized BIC Difference than HC across all tested sample sizes on Sachs, Alarm, and Win95pts. The largest observed score differences are 0.004 (Sachs, N=5000), 0.006 (Alarm, N=1000), and 0.007 (Win95pts, N=500).

(b) For GflowOpt on all datasets, the Normalized BIC Difference is often positive at smaller N. As N increases, the metric trends towards zero and can become negative, consistent with BIC's asymptotic properties.

(c) The rate of change in GflowOpt's Normalized BIC Difference diminishes as N increases. For instance, on Win95pts, the metric changes by 0.077 (N=500 to 1000) but only by 0.005 (N=10000 to 50000). This initial rate of change also appears correlated with network complexity (e.g., 0.077 for Win95pts vs. 0.029 for Alarm).

These results highlight GflowOpt's robust superiority over HC. The trends align with BIC's known complexity penalty and consistency properties [24].

5.2.3 Ablation. To quantitatively assess the contribution of key components, we perform ablation experiments on the Sachs dataset. Performance is measured by the mean Relative BIC score Improvement (normalized gain over an empty graph baseline) for four configurations presented in Table 4: the full GflowOpt, removing only Contrastive Loss (-CL), removing only Proxy Model guidance (-PM), and removing both (-CL-PM). The results show:

(a) The full GflowOpt configuration yields the highest mean Relative BIC score Improvement (22.08%).

(b) Removing only Contrastive Loss (-CL) or only Proxy Model guidance (-PM) reduces the mean improvement to 21.86% and 21.84%, respectively.

(c) Removing both components (-CL-PM) results in the lowest mean improvement (20.98%), corresponding to the largest performance decrease relative to the full configuration.

These results indicate that both the Contrastive Loss and the Proxy Model guidance make positive contributions to the BIC score improvement achieved by GFlowOpt, with their combined removal having the most significant negative impact.

5.3 Visualizations

5.3.1 Embedding Space Visualization. Figure 5 visually illustrates the impact of CL on the GFlowNet's representation space for DAGs on the Sachs dataset. It compares PCA-projected representations from GFlowNet models trained with versus without CL, for both complete states (fully formed DAGs) and intermediate states (partial DAGs). As visualized, CL leads to more tightly clustered representations for high-scoring final DAGs ((b) vs. (a)) and a smoother projection for random trajectories ((d) vs. (c)), an improvement that qualitatively supports the quantitative benefits of CL found in Table 4 for subsequent proxy model training.

5.3.2 Learned Structures Visualization. Figure 6 visually compares top BIC-scoring structures for the Asia dataset, with each displayed structure annotated with its BIC score and SHD to the ground truth. We can observe that:

Table 3: Comparison of average BIC scores for top-k structures found by DAG-GFlowNet and GFlowOpt across benchmark datasets. Best average score per column is bolded.

Method	Metric	Small Networks			Medium Networks			Large Networks		
		Asia	Sachs	Sports	Child	Alarm	Property	Hailfinder	Win95pts	ForMed
DAG-GFlowNet	Top-1	-11431±152	-38549±342	-56966±229	-70958±724	-98037±385	-198992±448	-348628±504	-95016±772	-299501±531
	Avg Top-10	-11433±142	-38592±148	-57052±149	-71165±423	-98593±408	-199594±407	-349225±497	-95818±754	-300302±505
GFlowOpt	Top-1	**-11266±109**	**-36502±120**	**-55606±105**	**-61976±308**	**-54522±302**	**-123332±178**	**-252288±257**	**-47212±309**	**-211171±389**
	Avg Top-10	**-11272±108**	**-36514±120**	**-55738±105**	**-62106±310**	**-54831±372**	**-123403±181**	**-252334±255**	**-47303±282**	**-211386±424**

Table 4: Ablation study results on Sachs. Performance measured by Relative BIC score Improvement.

Configuration	Rel. BIC Impr. (%)
Full GflowOpt	22.08 ± 0.35
No Contrastive Loss (-CL)	21.86 ± 0.32
No Proxy Model (-PM)	21.84 ± 0.34
No CL & No PM (-CL-PM)	20.98 ± 0.87

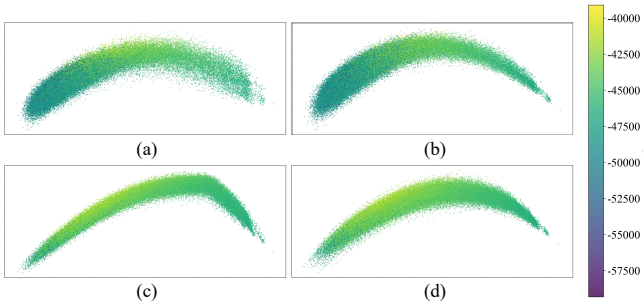

Figure 5: PCA visualization of GFlowNet DAG representations (Sachs dataset, N=5000), illustrating CL effects. Subfigure details – Models compared with CL vs. without CL, for representations of: (a) complete states, without CL; (b) complete states, with CL; (c) intermediate states, without CL; (d) intermediate states, with CL. Points are structure embeddings, colored by BIC score (lighter indicates higher scores).

(a) GFlowOpt's top unique structures for Asia exhibit low SHDs relative to the ground truth (values of 1, 2, and 5).

(b)These SHDs (1, 2, 5) are competitive with or superior to those from baselines (HC: SHD=2; PC-Max/MAHC: SHD=3; H2PC: SHD=5). Notably, GFlowOpt identifies a structure with SHD=1, outperforming these presented baselines.

This visual example underscores GFlowOpt's ability to discover structures with both high BIC scores and low SHD. Furthermore, identifying multiple distinct, high-quality structures suggests effective exploration of different modes and robust handling of model uncertainty.

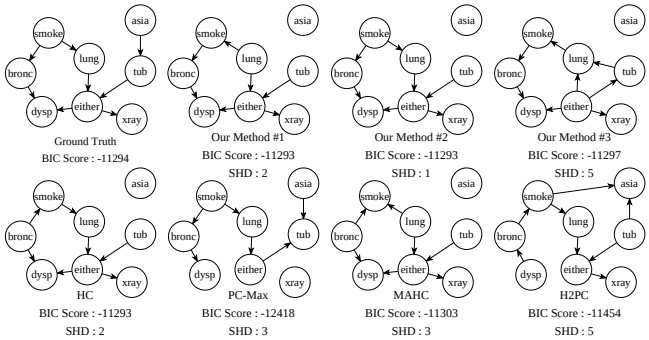

Figure 6: Top BIC-scoring graph structures for the Asia dataset (N=5000) from a representative run. Comparison includes: the Ground Truth network, GFlowOpt's Top-3 unique structures, and the single best structure from key baseline methods (HC, MAHC, H2PC, PC-MAX).

6 Conclusion

This paper introduced GFlowOpt, a novel framework that enhances GFlowNet-based posterior exploration with a two-stage, gradient-informed optimization strategy for discovering high-quality Bayesian Network structures. Our comprehensive experiments demonstrate that GFlowOpt consistently achieves superior BIC scores with reduced variability compared to state-of-the-art methods. This strong performance is attributed to its effective use of an improved GFlowNet, featuring BIC rewards and CL for enhanced representations, coupled with a robust proxy-guided continuous search and discrete HC refinement. While primarily optimizing for BIC score, our findings show that GFlowOpt also frequently identifies structures with high fidelity to the ground truth, underscoring its capability to produce accurate and trustworthy models.

Future research could focus on extending the GFlowOpt framework to learn BN structures with latent variables, and on exploring its application in diverse complex domains.

Acknowledgments

This work was supported by the Key Project of Fundamental Research of Yunnan Province of China (202501AS070102), the Joint Key Project of the National Natural Science Foundation of China (U23A20298), and the General Project of Fundamental Research of Yunnan Province of China (202401AT070462).

GenAI Usage Disclosure

The authors declare that no artificial intelligence content generation (AICG) technologies were utilized in the preparation of this work. All textual, analytical, and creative contributions were exclusively produced by the human authors through manual intellectual processes without automated assistance.

References

[1] Bruce Abramson, John Brown, Ward Edwards, Allan Murphy, and Robert L Winkler. 1996. Hailfinder: A Bayesian system for forecasting severe weather. *International Journal of Forecasting* 12, 1 (1996), 57–71.

[2] Ankur Ankan and Johannes Textor. 2024. pgmpy: A Python Toolkit for Bayesian Networks. *Journal of Machine Learning Research* 25, 265 (2024), 1–8.

[3] Yashas Annadani, Jonas Rothfuss, Alexandre Lacoste, Nino Scherrer, Anirudh Goyal, Yoshua Bengio, and Stefan Bauer. 2021. Variational causal networks: Approximate bayesian inference over causal structures. *arXiv preprint arXiv:2106.07635* (2021).

[4] Peter W Battaglia, Jessica B Hamrick, Victor Bapst, Alvaro Sanchez-Gonzalez, Vinicius Zambaldi, Mateusz Malinowski, Andrea Tacchetti, David Raposo, Adam Santoro, Ryan Faulkner, et al. 2018. Relational inductive biases, deep learning, and graph networks. *arXiv preprint arXiv:1806.01261* (2018).

[5] Ingo A Beinlich, Henri Jacques Suermondt, R Martin Chavez, and Gregory F Cooper. 1989. The ALARM monitoring system: A case study with two probabilistic inference techniques for belief networks. In *AIME 89: Second European Conference on Artificial Intelligence in Medicine*. 247–256.

[6] Emmanuel Bengio, Moksh Jain, Maksym Korablyov, Doina Precup, and Yoshua Bengio. 2021. Flow network based generative models for non-iterative diverse candidate generation. In *Advances in Neural Information Processing Systems 34: Annual Conference on Neural Information Processing Systems (NIPS)*. 27381–27394.

[7] Remco Ronaldus Bouckaert. 1995. *Bayesian belief networks: from construction to inference*. Ph. D. Dissertation. Utrecht University.

[8] John S Breese and David Heckerman. 1999. Decision-theoretic troubleshooting: A framework for repair and experiment. *Decision Science and Technology: Reflections on the Contributions of Ward Edwards* (1999), 271–287.

[9] David Maxwell Chickering. 1996. Learning Bayesian networks is NP-complete. In *Learning from Data - Fifth International Workshop on Artificial Intelligence and Statistics (AISTATS)*. 121–130.

[10] David Maxwell Chickering. 2002. Optimal structure identification with greedy search. *Journal of Machine Learning Research* 3, Nov (2002), 507–554.

[11] David Maxwell Chickering, David Heckerman, and Christopher Meek. 2004. Large-sample learning of Bayesian networks is NP-hard. *Journal of Machine Learning Research* 5, Oct (2004), 1287–1330.

[12] Diego Colombo, Marloes H Maathuis, et al. 2014. Order-independent constraint-based causal structure learning. *Journal of Machine Learning Research* 15, 1 (2014), 3741–3782.

[13] A Constantinou. 2019. The Bayesys user manual. *Queen Mary University of London* (2019).

[14] Anthony C Constantinou. 2022. Investigating the efficiency of the Asian handicap football betting market with ratings and Bayesian networks. *Journal of Sports Analytics* 8, 3 (2022), 171–193.

[15] Anthony C Constantinou and Norman Fenton. 2017. The future of the London buy-to-let property market: Simulation with temporal Bayesian networks. *Plos one* 12, 6 (2017), e0179297.

[16] Anthony Costa Constantinou, Mark Freestone, William Marsh, Norman Fenton, and Jeremy Coid. 2015. Risk assessment and risk management of violent reoffending among prisoners. *Expert Systems with Applications* 42, 21 (2015), 7511–7529.

[17] Anthony C Constantinou, Yang Liu, Neville K Kitson, Kiattikun Chobtham, and Zhigao Guo. 2022. Effective and efficient structure learning with pruning and model averaging strategies. *International Journal of Approximate Reasoning* 151 (2022), 292–321.

[18] Tristan Deleu, António Góis, Chris Emezue, Mansi Rankawat, Simon Lacoste-Julien, Stefan Bauer, and Yoshua Bengio. 2022. Bayesian structure learning with generative flow networks. In *Uncertainty in Artificial Intelligence, Proceedings of the Thirty-Eighth Conference on Uncertainty in Artificial Intelligence (UAI)*. 518–528.

[19] Tristan Deleu, Mizu Nishikawa-Toomey, Jithendaraa Subramanian, Nikolay Malkin, Laurent Charlin, and Yoshua Bengio. 2023. Joint bayesian inference of graphical structure and parameters with a single generative flow network. In *Advances in Neural Information Processing Systems 36: Annual Conference on Neural Information Processing Systems (NIPS)*. 31204–31231.

[20] Nir Friedman and Daphne Koller. 2003. Being Bayesian about network structure. A Bayesian approach to structure discovery in Bayesian networks. *Machine learning* 50 (2003), 95–125.

[21] Maxime Gasse, Alex Aussem, and Haytham Elghazel. 2014. A hybrid algorithm for Bayesian network structure learning with application to multi-label learning. *Expert Systems with Applications* 41, 15 (2014), 6755–6772.

[22] David Heckerman, Dan Geiger, and David M Chickering. 1995. Learning Bayesian networks: the combination of knowledge and statistical data. *Machine Learning* 20, 3 (1995), 197–243.

[23] Neville Kenneth Kitson, Anthony C Constantinou, Zhigao Guo, Yang Liu, and Kiattikun Chobtham. 2023. A survey of Bayesian Network structure learning. *Artificial Intelligence Review* 56, 8 (2023), 8721–8814.

[24] Daphne Koller and Nir Friedman. 2009. *Probabilistic graphical models: principles and techniques*. Vol. 26. MIT press.

[25] Jack Kuipers, Polina Suter, and Giusi Moffa. 2022. Efficient sampling and structure learning of Bayesian networks. *Journal of Computational and Graphical Statistics* 31, 3 (2022), 639–650.

[26] Steffen L Lauritzen and David J Spiegelhalter. 1988. Local computations with probabilities on graphical structures and their application to expert systems. *Journal of the Royal Statistical Society: Series B (Methodological)* 50, 2 (1988), 157–194.

[27] Lars Lorch, Jonas Rothfuss, Bernhard Schölkopf, and Andreas Krause. 2021. Dibs: differentiable Bayesian structure learning. In *Advances in Neural Information Processing Systems 34: Annual Conference on Neural Information Processing Systems (NIPS)*. 24111–24123.

[28] Renqian Luo, Fei Tian, Tao Qin, Enhong Chen, and Tie-Yan Liu. 2018. Neural architecture optimization. In *Advances in Neural Information Processing Systems 31: Annual Conference on Neural Information Processing Systems (NIPS)*.

[29] David Madigan, Jeremy York, and Denis Allard. 1995. Bayesian graphical models for discrete data. *International Statistical Review/Revue Internationale de Statistique* (1995), 215–232.

[30] Judea Pearl. 2014. *Probabilistic reasoning in intelligent systems: networks of plausible inference*. Elsevier.

[31] Joseph Ramsey. 2016. Improving accuracy and scalability of the PC algorithm by maximizing p-value. *arXiv preprint arXiv:1610.00378* abs/1610.00378 (2016).

[32] Karen Sachs, Omar Perez, Dana Pe'er, Douglas A Lauffenburger, and Garry P Nolan. 2005. Causal protein-signaling networks derived from multiparameter single-cell data. *Science* 308, 5721 (2005), 523–529.

[33] M. Scutari. 2010. Learning Bayesian Networks with the bnlearn R Package. *Journal of Statistical Software* 35, 3 (2010), 1–22.

[34] Moninder Singh and Marco Valtorta. 2013. An Algorithm for the Construction of Bayesian Network Structures from Data. *CoRR* abs/1303.1485 (2013).

[35] David J Spiegelhalter. 1992. Learning in probabilistic expert systems. *Bayesian statistics* 4 (1992), 447–465.

[36] Peter Spirtes and Clark Glymour. 1991. An algorithm for fast recovery of sparse causal graphs. *Social Science Computer Review* 9, 1 (1991), 62–72.

[37] Peter Spirtes, Clark N Glymour, and Richard Scheines. 2000. *Causation, prediction, and search*. Vol. 45. MIT press.

[38] Ioannis Tsamardinos, Laura E Brown, and Constantin F Aliferis. 2006. The max-min hill-climbing Bayesian network structure learning algorithm. *Machine Learning* 65, 1 (2006), 31–78.

[39] Jussi Viinikka, Antti Hyttinen, Johan Pensar, and Mikko Koivisto. 2020. Towards scalable bayesian learning of causal dags. *Advances in Neural Information Processing Systems* 33 (2020), 6584–6594.

[40] Chirayu (Kong) Wongchokprasitti, Alexander Murray-Watters, Jeremy Espino, and Roger Day. 2019. *bd2kccd/r-causal: v1.2.1*. https://doi.org/10.5281/zenodo.3592990

[41] Yuning You, Tianlong Chen, Yongduo Sui, Ting Chen, Zhangyang Wang, and Yang Shen. 2020. Graph Contrastive Learning with Augmentations. (2020).